Trade-offs, Comparative Advantage, and the Market System

ECON 101 - Macroeconomics

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Chapter Outline

- **2.1** Production Possibilities Frontiers and Opportunity Costs
- **2.2** Comparative Advantage and Trade
- **2.3** The Market System

Scarcity and Trade-offs

- People continually face decisions about how best to use their **scarce** resources.
- **Scarcity:** a situation in which unlimited wants exceed the limited resources available to fulfill those wants.
- Scarcity requires **trade-offs**: to get more of one thing, society must accept less of another.
- Economics gives us tools to help make these trade-offs wisely.

Example: Tesla and Trade-offs

- Tesla has scarce workers and machinery.
- If Tesla wants to produce more Model 3 sedans, those resources are not available to produce Cybertruck pickups (and vice versa).
- This illustrates the central macro idea: even a high-tech firm cannot produce unlimited quantities of everything at once.

2.1 Learning Objective

Goal

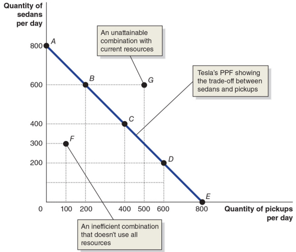
Use a production possibilities frontier to analyze opportunity costs and trade-offs.

- A **production possibilities frontier (PPF)** is a curve showing the maximum attainable combinations of two goods that can be produced with available resources and current technology.
- The PPF is a **positive** tool: it shows what *is* possible, not what *should* be produced.

Figure 2.1 Tesla's Production Possibilities Frontier

Tesla faces a trade-off: To build one more sedan, it must build one less pickup. The PPF illustrates the trade-off Tesla faces. Combinations on the PPF—such as points *A*, *B*, *C*, *D*, and *E*—are *productively efficient* because the maximum output is being obtained from the available resources. Combinations inside the frontier—such as point *F*—are *inefficient* because some resources are not being used. Combinations outside the frontier—such as point *G*—are *unattainable* with current resources.

Tesla's Production Possibilities at Its Plant		
Choice	Quantity of Model 3 Sedans Produced	Quantity of Cybertruck Pickups Produced
A	800	0
B	600	200
C	400	400
D	200	600
E	0	800
F	300	100
G	600	500



Interpreting the PPF: Tesla Example

- Tesla can produce combinations of Model 3 sedans and Cybertruck pickups.
- **Points on** the PPF: attainable and **productive efficient**.
- **Points below** the PPF: attainable but **inefficient** (resources underused).
- **Points above** the PPF: currently unattainable with existing resources and technology.

Figure 2.1 – Tesla’s PPF and Opportunity Cost (2 of 2)

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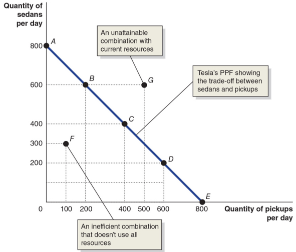


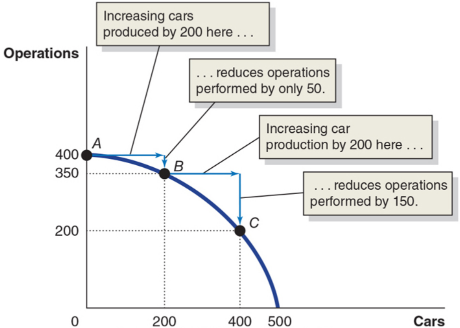
Opportunity Cost on the PPF

- Suppose moving from point A to point B:
  - Tesla produces 200 more Cybertruck pickups.
  - To do so, it must produce 200 fewer Model 3 sedans.
- The **opportunity cost** of producing 200 more Cybertrucks is the 200 Model 3 sedans given up.
- **Opportunity cost:** the highest-valued alternative that must be given up to engage in an activity.

Figure 2.2 – Increasing Marginal Opportunity Costs

Figure 2.2 Increasing Marginal Opportunity Costs

As the economy moves down the PPF, it experiences *increasing marginal opportunity costs* because increasing car production by a given quantity requires larger and larger decreases in the number of operations performed. For example, to increase car production from 0 to 200—moving from point *A* to point *B*—the economy has to give up only 50 operations. But to increase production of cars by another 200 vehicles—moving from point *B* to point *C*—the economy has to give up 150 operations.

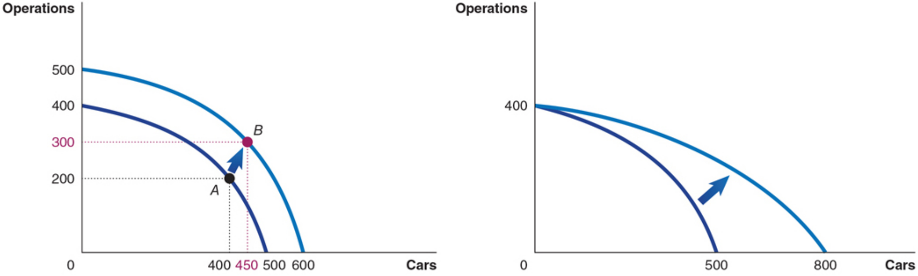


Why Opportunity Costs Often Increase

- Some resources are better suited to one task than another.
- The first resources to be switched are the ones best suited to switching.
- As more resources are devoted to an activity, the extra output from additional resources tends to fall.
- This gives the PPF its bowed-out shape and leads to **increasing marginal opportunity cost**.

Figure 2.3 Economic Growth

Panel (a) shows that as more economic resources become available and technological change occurs, the economy can move from point *A* to point *B*, performing more operations and producing more cars. Panel (b) shows the results of technological change in the automobile industry that increases the quantity of cars workers can produce per year while leaving unchanged the maximum quantity of operations that can be performed. Shifts in the PPF represent *economic growth*.



(a) Shifting out the PPF **(b) Technological change in the automotive industry**

Economic Growth and the PPF

- As more economic resources become available, the economy can move from point A to point B, producing more of both goods.
- Shifts in the PPF represent **economic growth**.
- **Economic growth:** the ability of the economy to increase the production of goods and services.
- Macro link: long-run growth allows higher living standards without giving up as much of other goods.

Figure 2.3 – Technological Change in One Sector

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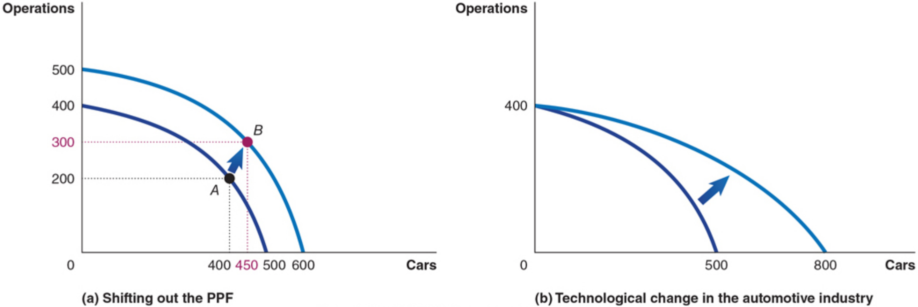

Technological Change in One Industry

- Suppose technology improves in the automobile industry only.
- The maximum output of automobiles increases, but the production possibility of the other good (e.g., tanks) is unchanged.
- Previously unattainable combinations become attainable.
- This captures sector-specific technological progress common in macroeconomics (e.g., ICT revolution).

2.2 Learning Objective

Goal

Understand comparative advantage and explain how it serves as the basis for trade.

Two-Person Production Example: You and Your Neighbour

- You and your neighbour each have limited time to pick apples and/or cherries.
- The table (in the slides) shows:
  - If you spend all your time picking cherries, you can pick 20 kg of cherries.
  - If you spend all your time picking apples, you can pick 20 kg of apples.
  - Your neighbour can pick 60 kg of cherries or 30 kg of apples if she specializes.
- We use these numbers to illustrate absolute and comparative advantage.

Figure 2.4 – Production Possibilities for You and Your Neighbour, Without Trade

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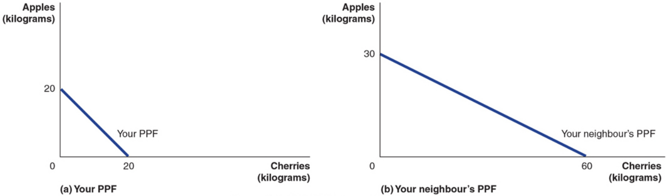
The table in this figure shows how many kilograms of apples and how many kilograms of cherries you and your neighbour can each pick in one week. The graphs in the figure use the data from the table to construct PPFs for you and your neighbour. Panel (a) shows your PPF. If you devote all your time to picking apples and none of it to picking cherries, you can pick 20 kilograms. If you devote all your time to picking cherries, you can pick 20 kilograms. Panel (b) shows that if your neighbour devotes all their time to picking apples, they can pick 30 kilograms. If they devote all their time to picking cherries, they can pick 60 kilograms.

	You		Your Neighbour	
	Apples (kg)	Cherries (kg)	Apples (kg)	Cherries (kg)
Pick nothing but apples	20	0	30	0
Pick nothing but cherries	0	20	0	60



Autarky (No Trade) Outcomes

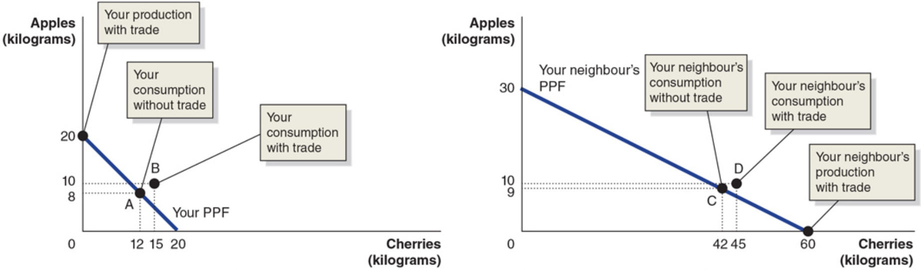
- Without trade, each person consumes only what they produce.
- **You:** pick and consume 8 kg of apples and 12 kg of cherries per week (point A).
- **Neighbour:** picks and consumes 9 kg of apples and 42 kg of cherries per week (point C).
- Both are on their own PPFs but may not be at the best possible consumption points if trade is allowed.

Specialization and Gains from Trade

- **Trade:** the act of buying and selling.
- What if you and your neighbour decide to specialize and trade?
- Your neighbour appears better at picking both apples and cherries, but:
  - You can still both benefit from trade by specializing in what you are **relatively** good at.

Figure 2.5 Gains from Trade

When you don't trade with your neighbour, you pick and consume 8 kilograms of apples and 12 kilograms of cherries per week—point *A* in panel (a). When you neighbour doesn't trade with you, they pick and consume 9 kilograms of apples and 42 kilograms of cherries per week—point *C* in panel (b). If you specialize in picking apples, you can pick 20 kilograms. If your neighbour specializes in picking cherries, they can pick 60 kilograms. If you trade 10 kilograms of apples for 15 kilograms of your neighbour's cherries, you will be able to consume 10 kilograms of apples and 15 kilograms of cherries—point *B* in panel (a). You neighbour can now consume 10 kilograms of apples and 45 kilograms of cherries—point *D* in panel (b). You and your neighbour are both better off as a result of the trade.



(a) Your production and consumption with trade

(b) Your neighbour's production and consumption with trade

Numerical Trade Example

- If you specialize in picking apples, you can pick 20 kg.
- If your neighbour specializes in picking cherries, she can pick 60 kg.
- Suppose you trade 10 kg of your apples for 15 kg of your neighbour's cherries:
  - **You** consume 10 kg of apples and 15 kg of cherries (point B).
  - **Neighbour** consumes 10 kg of apples and 45 kg of cherries (point D).
- Both are better off than in autarky.

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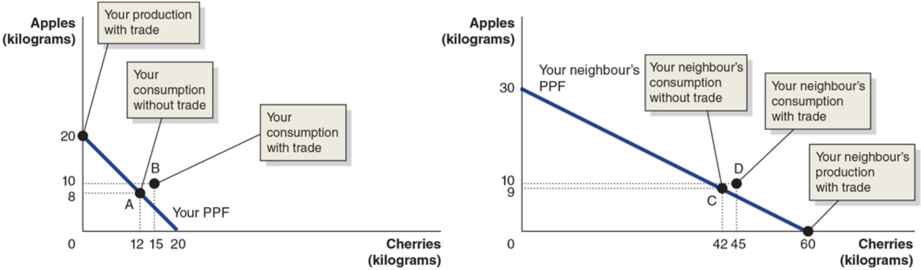

(a) Your production and consumption with trade **(b) Your neighbour's production and consumption with trade**

Table 2.1 – Summary of the Gains From Trade

Table 2.1 A Summary of the Gains from Trade

	You		Your Neighbour	
	Apples (kg)	Cherries (kg)	Apples (kg)	Cherries (kg)
Production and consumption *without* trade	8	12	9	42
Production *with* trade	20	0	0	60
Consumption *with* trade	10	15	10	45
Increased consumption (gains from trade)	2	3	1	3

Table 2.1 – Summary of the Gains From Trade

- Both you and your neighbour are able to consume more with trade than without trade.
- Key points to emphasize to students:
  - Trade allows consumption points outside each individual's own PPF.
  - Even if one person is better at producing *both* goods, both can gain.

Explaining the Gains from Trade

- Your neighbour has an **absolute advantage** in both cherry and apple picking.
- You have a **comparative advantage** in picking apples.
- **Absolute advantage:** ability to produce more of a good or service than competitors using the same amount of resources.
- **Comparative advantage:** ability to produce a good or service at a lower opportunity cost than competitors.

Table 2.2 – Opportunity Costs of Picking Apples and Cherries

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	Opportunity Cost of Apples	Opportunity Cost of Cherries
You	1 kg of cherries	1 kg of apples
Your neighbour	2 kg of cherries	0.5 kg of apples

Table 2.2 – Opportunity Costs of Picking Apples and Cherries

- This table compares the opportunity cost of apples vs. cherries for you and your neighbour.
- The key lesson:
  - The basis for trade is **comparative advantage**, not absolute advantage.
  - Individuals, firms, and countries are better off if they specialize in producing goods and services for which they have a comparative advantage and obtain the other goods and services they need by trading.

2.3 Learning Objective

Goal

Explain the basic idea of how a market system works.

Markets and the Modern Economy

- A **market** is a group of buyers and sellers of a good or service, and the institution or arrangement by which they come together to trade.
- Two key groups participate in the modern economy:
  - **Households**
  - **Firms**

Households and Firms

- **Households** consist of individuals who provide the **factors of production**:
  - labour, capital, natural resources, and entrepreneurial ability.
- Households receive payments for these factors by selling them to firms in **factor markets**.
- **Firms** supply goods and services to **product markets**; households buy these products from the firms.

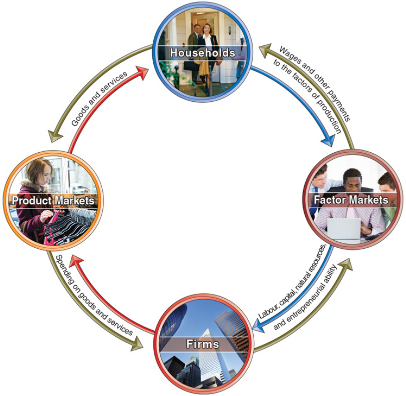
The Four Factors of Production

- **Labour:** all types of work, from part-time jobs at McDonaldś to CEOs of large corporations.
- **Capital:** physical capital such as computers, machines, and buildings used to produce other goods.
- **Natural resources:** land, water, oil, iron ore, and other raw materials ("gifts of nature").
- **Entrepreneur:** someone who operates a business.
- **Entrepreneurial ability:** ability to bring together the other factors of production to produce and sell goods and services successfully.

Figure 2.6 – The Circular-Flow Diagram (1 of 2)

Figure 2.6 A Simple Circular-Flow Diagram

Households and firms are linked together in a circular flow of production, income, and spending. The blue arrows show the flow of the factors of production. In factor markets, households supply labour, entrepreneurial ability, and other factors of production to firms. Firms use these factors of production to make goods and services that they supply to households in product markets. The red arrows show the flow of goods and services from firms to households. The green arrows show the flow of funds. In factor markets, households receive wages and other payments from firms in exchange for supplying the factors of production. Households use these wages and other payments to purchase goods and services from firms in product markets. Firms sell goods and services to households in product markets, and they use the funds to purchase the factors of production from households in factor markets.



Circular-Flow Diagram: Basic Idea

- The **circular-flow diagram** is a model that illustrates how participants in markets are linked.
- **Real flows:**
  - Households provide factors of production to firms.
  - Firms provide goods and services to households.
- **Money flows:**
  - Firms pay money to households for the factors of production.
  - Households pay money to firms for goods and services.

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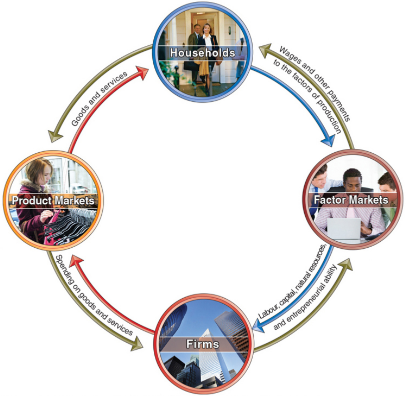

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Limits of the Simple Circular-Flow Model

- Like all economic models, the circular-flow diagram is a **simplified** version of reality.
- The basic version excludes:
  - Government
  - Financial system (banks, credit markets)
  - Foreign buyers and sellers of goods and services
- We add these sectors later in the course to build full macro models.

Gains from Free Markets

- A **free market** has few government restrictions on:
  - how goods or services can be produced or sold, or
  - how factors of production can be employed.
- Countries that come closest to this benchmark have typically been more successful than centrally planned economies in raising living standards.
- Classic idea: Adam Smithś **"invisible hand"** – individuals acting in their own self-interest collectively satisfy the wants of consumers.

Things to Remember

- The production possibilities frontier should **never bow inward**. Why?
  - Because opportunity cost typically increases as you move along the PPF.
- The PPF tells us what **can** be produced, not what **should** be produced.
- Just because someone is better or worse at *everything* does not mean trade with them cannot be beneficial.
- The basis for trade is **comparative advantage**, not absolute advantage.
- Free markets raise the standard of living, but that does not mean there is no role for governments:
  - Governments must provide a sound legal environment for the market system to succeed.

Summary

- PPFs illustrate trade-offs, opportunity cost, economic growth, and technological change.
- Comparative advantage explains why individuals and countries gain from specialization and trade, even when one side has an absolute advantage in everything.
- The market system coordinates decisions of households and firms through product and factor markets.
- The circular-flow diagram provides a simple macro picture of how goods, services, and money move in the economy.
- Free markets and well-functioning institutions are central to long-run improvements in living standards.

Thank you!